

A Connectome Test of the Fly Hashing Algorithm

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Abstract—Dasgupta, Stevens and Navlakha (2017) described the *Drosophila* olfactory circuit as a locality-sensitive hash whose glomerulus-to-Kenyon-cell projection was drawn at random because the wiring was unknown. We test that idealisation against four connectomes, MaleCNS, hemibrain, FlyWire and BANC: four animals, seven hemispheres. We reimplement the 2017 benchmark protocol (SIFT, GloVe, MNIST, plus odour mixtures) and recover its central pattern: with random matrices the fly hash beats k Gaussian projections at short hash lengths ($3.1\times$ in average precision on MNIST at $k = 4$). Substituting the connectome keeps that advantage, and retrieval stays within a few percent of uniform rewirings that preserve both degree sequences, slightly below them at large hash sizes. At equal connection count, spreading each glomerulus’s output evenly over Kenyon cells improves retrieval by up to +18%; equalising inputs per cell does not. On DoOR odour mixtures the MaleCNS connectome scores -2.06% below its rewirings (two-sided $p = 0.020$), a deficit that disappears when unmeasured responses are imputed. Finally, the 2017 advantage is per active cell: real-valued Gaussian projections given the same projection arithmetic as the fly network beat it on every dataset.

Index Terms—connectomics, locality-sensitive hashing, similarity search, replication

I. INTRODUCTION

Locality-sensitive hashing (LSH) maps items to short codes under which similar items collide with high probability [1], [2]. Dasgupta *et al.* [3] noticed that the fly olfactory circuit computes a related transform with three unusual features: a sparse binary projection, an expansion rather than a reduction of dimension, and winner-take-all. On SIFT, GloVe and MNIST it beat classical LSH by up to about $3\times$. The projection was sampled at random, following the account that this wiring is effectively random [4].

That account has since been revised: projection-neuron input to Kenyon cells is structured [5]. Complete projections are now available from four electron-microscopy connectomes [6], [10], [11], [13]. We ask whether the random matrix was a harmless idealisation for the task that motivated it, in two ways:

- *Approach 1*: reimplement the 2017 protocol, check it against the published pattern, add odour mixtures as a fourth dataset, and substitute each connectome for the random matrix.
- *Approach 2*: an odour analysis adapted to the circuit, with randomization tests, a bootstrap over odorant sets and an equivalence margin.

Both compare the connectome with the same null, with controls that change one degree sequence at a time at equal connection count, and with LSH under three budgets. To our

knowledge no prior work has run the 2017 benchmark on a connectome; [5] studied odour discrimination, not retrieval.

II. COMMON DESIGN

A. Connectomes

Let $M \in \{0, 1\}^{d \times m}$ with $M_{ji} = 1$ iff glomerulus j connects to Kenyon cell i . For every connectome the same rule applies: uniglomerular olfactory projection neurons are assigned to glomeruli by cell type (multiglomerular, thermo- and hygro-sensory types excluded), Kenyon cells are those annotated as such on the same side, every reconstructed connection counts (no synapse threshold), and cells without olfactory input are dropped. Hemibrain glomerulus names are mapped to the current nomenclature using the cross-dataset matches of [12]. We use MaleCNS v1.0 (male, both hemispheres), hemibrain v1.2 (female, right only), FlyWire v783 and BANC v888 (female, both). All hemispheres share 49 glomeruli; they have 1,761 to 2,406 Kenyon cells with 5.00 to 6.05 glomerular inputs per cell on average. The datasets differ in imaging, synapse detection and proofreading, so part of any difference between them is technical. The primary connectome is the MaleCNS right hemisphere ($d = 51$, $m = 1,886$; 5.50 distinct projection-neuron partners per cell).

B. Hash, null and controls

The fly hash computes $y = M^\top x$ and sets the k largest entries of y to one. Only M changes between conditions. *The null* is uniform over binary matrices with the same row and column sums as M , sampled by curveball trades [7], [8]; it randomises which glomeruli share a cell and keeps both degree sequences. We use 30 trades per cell; co-occurrence $Q = \sum_{j < l} (MM^\top)_{jl}^2$ stays within three null SDs from 1 sweep on. Tests against it are approximate Monte Carlo randomization tests.

Equal-connection controls have exactly $\text{nnz}(M)$ ones and change one margin at a time: inputs per cell made as even as possible (fan-out kept), fan-out per glomerulus made even (inputs kept), or both; the pairing is then randomised by curveball. *The 2017 construction* gives every cell six uniform inputs; it changes both margins and the connection count.

C. Scores, baselines and budgets

Retrieval is scored by average precision at n , the number of true neighbours: $\text{AP}@n = \frac{1}{n} \sum_i \text{prec}@i \cdot \text{rel}(i)$, where true neighbours never retrieved count as misses; $\text{recall}@n$ is reported with it. LSH computes k Gaussian projections kept as real values, as in the 2017 comparison, or their signs for bit codes. Budgets are *per active cell* (k projections for k

TABLE I

PROVENANCE OF THE PROTOCOL: FROM THE 2017 MAIN TEXT [3], FROM THE 2018 FLYLSH CODE [14], OR CHOSEN HERE.

element	choice	source
data	10,000 vectors: SIFT-small base, 2017 (size); ours GloVe 6B 300d and MNIST train (subset)	
queries, truth	1,000 queries; top 2% by Euclidean distance on raw features	2017
preprocessing	each vector centred by its own mean	2017
fly	$m = 20k$ or $10d$; top- k winners	2017
sampling	each cell samples 10% of inputs	2018 code
tag	binary; Hamming distance, random tie-break	2018 code; ours
LSH	k real-valued Gaussian projections	2017
score	AP@200 and recall@200	ours (2017: undefined “mAP”)
trials	50; matrices and queries redrawn	2017

TABLE II

RANDOM MATRICES, 2017 PROTOCOL: AP@200 (RECALL@200), MEAN OF 50 TRIALS.

data	d	k	LSH	rand. sel.	fly 20 k	fly 10 d	10 d /LSH
SIFT	128	4	0.033 (0.13)	0.034	0.064	0.093 (0.22)	2.8×
SIFT	128	16	0.198 (0.35)	0.199	0.245	0.269 (0.41)	1.4×
GloVe	300	4	0.002 (0.03)	0.002	0.004	0.005 (0.05)	2.2×
GloVe	300	16	0.016 (0.10)	0.016	0.016	0.015 (0.08)	1.0×
MNIST	784	4	0.031 (0.12)	0.034	0.062	0.094 (0.20)	3.1×
MNIST	784	16	0.221 (0.35)	0.221	0.253	0.276 (0.40)	1.2×
Odours	35	4	0.094 (0.22)	0.093	0.128	0.159 (0.29)	1.7×
Odours	35	16	0.393 (0.50)	0.380	0.287	0.290 (0.42)	0.7×

winners), *per stored bit* and *per operation*. Operations are the projection arithmetic of [3]: d multiplications and d additions per Gaussian projection, one addition per non-zero of M ; winner selection, retrieval and any PCA fit are excluded.

III. APPROACH 1: THE 2017 PROTOCOL

A. Reimplementation

Table I lists each protocol element and where it comes from. The 2017 paper reports “mean average precision” without a formula. Precision averaged over *retrieved* true neighbours only, a convention in parts of the hashing literature, reproduces its LSH baseline on MNIST at $k = 4$ (0.159 against 0.160) but not its fly score (0.365 against 0.448). That score rewards retrieving a few easy neighbours (one hit at rank 1 scores 1), so we use AP@200 and recall@200 and compare with the paper only through orderings and ratios. The FlyLSH code computes neighbours on centred rather than raw input; scoring against those leaves the MNIST ratio at 3.1×

B. Random matrices

The published pattern holds (Table II). Random tag selection performs like LSH, as the paper argues it must; winner-take-all at $m = 20k$ beats both; the 10 d expansion does better still; the advantage is largest at short hash lengths (3.1×

on MNIST at $k = 4$; recall 1.7×

TABLE III

INPUT DIMENSION ALONE: MNIST REDUCED BY PCA TO d COMPONENTS, $k = 4, 10$ TRIALS. OPS: 10 d FLY OVER 4-PROJECTION LSH. LAST THREE COLUMNS: LSH WITH THE 10 d FLY’S ARITHMETIC, ITS PROJECTIONS AND AP, AND THE FLY’S BEST AP OVER $k \leq 16$.

d	ops	LSH	fly 10 d	ratio	proj.	LSH eq. ops	fly best
8	1×	0.172	0.053	0.3×	5	0.216	0.083
16	2×	0.109	0.110	1.0×	10	0.323	0.229
32	4×	0.070	0.143	2.1×	15	0.368	0.293
64	8×	0.056	0.130	2.3×	30	0.464	0.307
128	16×	0.041	0.118	2.8×	65	0.591	0.294
256	32×	0.040	0.110	2.8×	130	0.689	0.277
512	64×	0.033	0.104	3.2×	255	0.772	0.264

TABLE IV

MALECONS CONNECTOME, 2017 PROTOCOL, AP@200: CONNECTOME, NULL MEAN ($B = 50$), RELATIVE DIFFERENCE, TWO-SIDED p (FLOOR 0.039), THE SIX-INPUT CONSTRUCTION AND LSH WITH k PROJECTIONS.

data	k	conn.	null	$\Delta\%$	p	six	LSH
SIFT	4	0.101	0.101	+0.0	0.941	0.104	0.043
SIFT	16	0.234	0.233	+0.4	0.431	0.236	0.252
SIFT	94	0.381	0.387	-1.7	0.039	0.448	0.642
GloVe	4	0.022	0.022	+0.0	0.980	0.025	0.008
GloVe	16	0.066	0.066	-0.1	0.941	0.070	0.081
GloVe	94	0.153	0.159	-3.5	0.039	0.182	0.431
MNIST	4	0.144	0.144	-0.2	0.784	0.155	0.058
MNIST	16	0.322	0.325	-0.8	0.118	0.329	0.327
MNIST	94	0.475	0.482	-1.6	0.039	0.517	0.711
Odours	4	0.158	0.166	-4.9	0.235	0.163	0.096
Odours	16	0.295	0.300	-1.5	0.275	0.309	0.396
Odours	92	0.362	0.375	-3.5	0.039	0.409	0.647

hash trails LSH at $d = 8$, ties at 16 and leads from 32 on. At equal projection arithmetic real-valued LSH leads at every d .

C. The connectome

M accepts only its own glomeruli as input. SIFT, GloVe and MNIST are reduced by PCA to 51 components; since image and word features have no glomerulus, each trial assigns components to glomeruli by a fresh random permutation, so these datasets test the wiring’s statistics, not its odour-specific pairing. Odours use the 35 glomeruli DoOR covers ($m = 1,838$). Over 20 trials we score M , 50 nulls, one draw of each control per trial, and LSH.

The connectome keeps the fly hash’s advantage over LSH at short hash lengths: 0.144 against 0.058 on MNIST at $k = 4$ (2.5×

1.6×

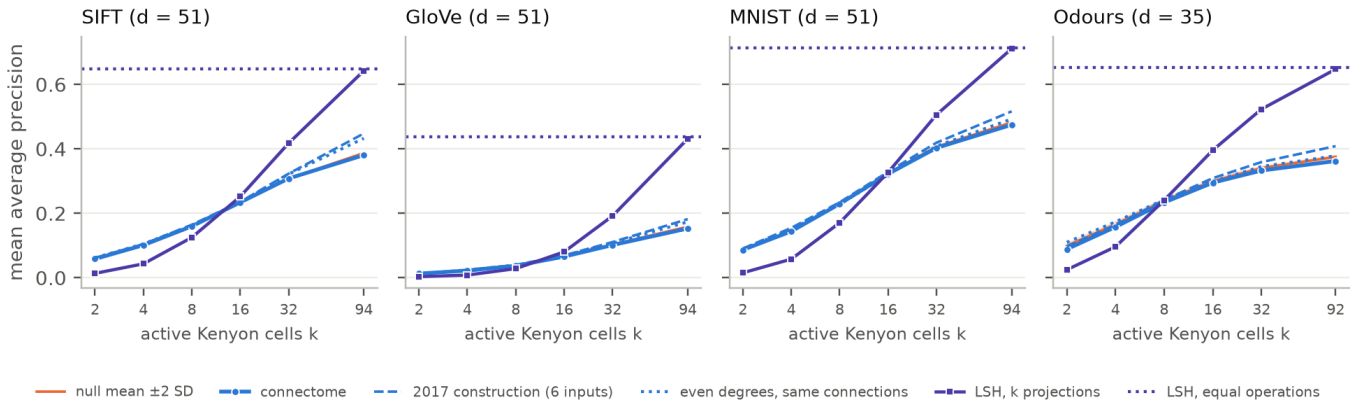


Fig. 1. The 2017 protocol with the MaleCNS connectome (right hemisphere), AP@200. SIFT, GloVe and MNIST are reduced by PCA to 51 components assigned to glomeruli at random in each trial; odours use the 35 measured glomeruli. All methods see the same input. The null band is mostly too narrow to see.

TABLE V
DEGREE CONTROLS, MALECNS, RELATIVE TO THE NULL MEAN (%).
EQUAL CONNECTIONS: INPUTS PER CELL EVEN, FAN-OUT EVEN, BOTH
EVEN; THE SIX-INPUT CONSTRUCTION CHANGES THE CONNECTION
COUNT. RIGHT: THE SAME WITHOUT ROW-CENTRING (CONNECTOME,
BOTH EVEN).

data	k	equal connections					no centring	
		conn.	in	out	both	six	conn.	both
SIFT	4	+0.0	-2.4	+4.6	+3.9	+3.2	+0.3	+6.2
SIFT	94	-1.7	-5.8	+15.9	+11.7	+15.6	-1.7	+13.5
GloVe	4	+0.0	+5.8	+4.8	+10.2	+11.4	-0.1	+12.8
GloVe	94	-3.5	-6.8	+17.7	+9.5	+14.8	-3.8	+7.5
MNIST	4	-0.2	+0.1	+3.1	+4.1	+7.2	-0.3	+5.1
MNIST	94	-1.6	-5.4	+7.8	+2.2	+7.1	-1.6	+2.7
Odours	4	-4.9	+5.5	+2.0	+4.7	-1.5	-11.7	+77.0
Odours	92	-3.5	-2.9	+3.2	+0.9	+8.9	-2.7	-6.5

many inputs a cell has. What the circuit gives up relative to idealised wiring is thus mainly its uneven use of glomeruli, not its pairing.

IV. APPROACH 2: ODOUR ANALYSIS

A. Design

Responses are the DoOR 2.0 consensus matrix [9] (commit db323a4). Coverage is uneven: 35 glomeruli are measured for 51 or more odorants, eleven for 12 to 24, and five for none. We keep the 35 (a threshold of 40 odorants, in that gap; our choice) and odorants measured on at least 12 of them: 172 odorants, 28.7% of entries unmeasured and set to zero. The benchmark is 4,000 synthetic linear mixtures of two to five profiles, normalised to unit mean as in the circuit; ground truth is the $\kappa = 10$ nearest items on raw responses; AP is computed over all κ neighbours. The restricted connectome has mean 3.87 inputs and 4.04 projection-neuron partners per cell.

The primary size $k = 92$ (5% of cells) and the family of secondary sizes $\{8, 16, 32, 64\}$ (Holm-adjusted) were fixed in an earlier revision of this analysis, after first results had been seen; this is not a preregistration. We compare M with

TABLE VI
ODOUR ANALYSIS, MALECNS: CONNECTOME AGAINST THE NULL
($B = 200$) AND COMPARISON MATRICES (RELATIVE TO THE NULL MEAN,
%). *PRIMARY.

k	conn.	null	$\Delta\%$	p	p_{Holm}	fan	flat	6 unif.	balanced
8	0.1973	0.2034 $\pm$ 0.0085	-3.0	0.468	1.000	+3.7	+26.7	+3.2	
16	0.2879	0.2933 $\pm$ 0.0062	-1.8	0.438	1.000	+2.7	+9.7	+3.2	
32	0.3531	0.3575 $\pm$ 0.0045	-1.2	0.338	1.000	+3.3	+3.0	+3.6	
64	0.3963	0.4001 $\pm$ 0.0039	-0.9	0.348	1.000	+3.4	+0.7	+3.4	
92*	0.4063	0.4148 $\pm$ 0.0035	-2.1	0.020	—	+3.2	+0.1	+3.2	

$B = 200$ nulls. A two-stage bootstrap resamples source odorants, regenerates the benchmark and draws 20 nulls without replacement from the same pool of 200 (200 replicates); its coverage was checked for one fitted rank-6 generator at 600 items. Equivalence is assessed at $\pm 5\%$.

B. Results

The pairing is structured: glomerulus co-occurrence Q is +3.7% above the null mean and exceeds all 200 draws ($p = 0.005$), as in [5]. Retrieval at the primary size is 0.4063 against 0.4148 ± 0.0035 , a relative difference of -2.06% (two-sided $p = 0.020$ unadjusted; 0.100 if all five sizes are treated as one family; Table VI, Fig. 2). Secondary sizes give -3.0% to -0.9%. The bootstrap 90% interval is -3.1% to -0.8%; equivalence at $\pm 5\%$ holds, at $\pm 2.5\%$ it does not hold. The interval covered the true value in 100% of 100 simulated datasets at nominal 90%, for that one generator. The two-sided test has 80% power against shifts of 2.5% downward and 3.0% upward.

Across 26 one-factor variations (Table VII) the difference ranges from -3.9% to +0.8% and is negative in 24. The deficit depends on the input: imputing unmeasured responses gives -0.7% to +0.6% ($p \geq 0.439$), and all 46 glomeruli with data give no difference. It cannot be separated from the treatment of unmeasured responses and the choice of glomeruli.

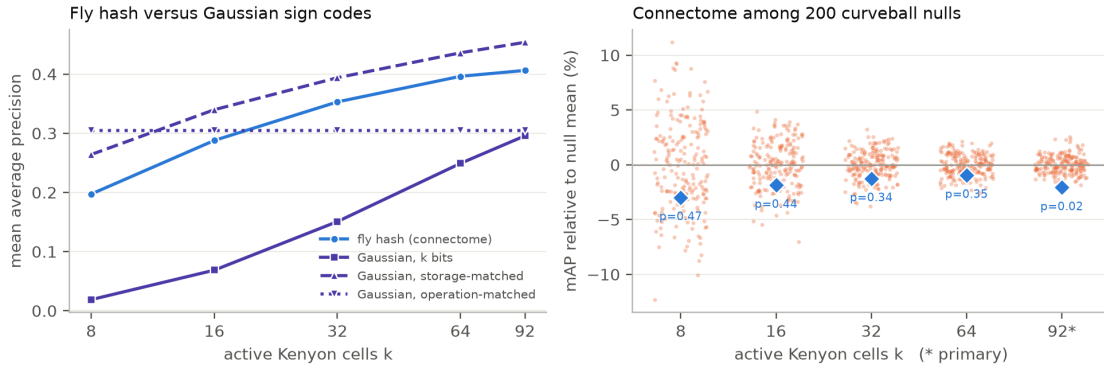


Fig. 2. Odour analysis, MaleCNS. **Left:** AP of the fly hash on the connectome and of Gaussian sign codes with k bits, at matched storage and at matched operations (102 bits); mean of 20 seeds. **Right:** each orange point is one of 200 null matrices, blue diamonds the connectome, both relative to the null mean.

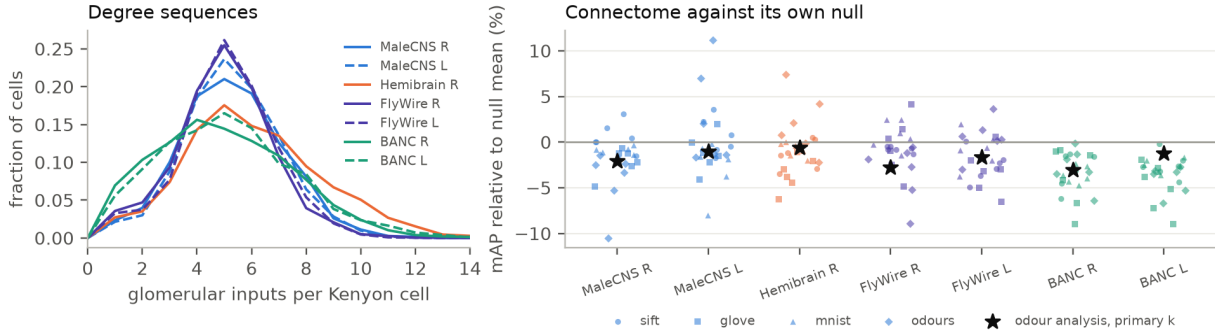


Fig. 3. Four connectomes. **Left:** glomerular inputs per Kenyon cell. **Right:** each hemisphere’s relative difference to its own null, per dataset and hash size under the 2017 protocol (small markers) and at the primary size of the odour analysis (stars).

TABLE VII

ODOUR ANALYSIS, MALECNS, PRIMARY SIZE, ONE-FACTOR VARIATIONS ($B = 40$ NULLS EACH): RELATIVE DIFFERENCE (%) AND TWO-SIDED p .

variation	$\Delta\%$	p
Baseline	-2.2	0.049
Left hemisphere	-1.1	0.244
≥ 5 -synapse threshold	-2.3	0.098
Missing data (3 imputations)	-0.7 to +0.6	0.439–0.780
Coverage thresholds (2)	-1.7 to -1.4	0.049–0.341
All 46 glomeruli with data, imputed	-0.0	1.000
Mixture seeds (4)	-2.4 to -1.6	0.049–0.244
Mixture size 2 / 2–10	-2.0 to -0.9	0.146–0.195
Measured odorants only	-3.9	0.049
Normalised / angular metric	-2.4 to -1.7	0.049
Tie rules (index, include, 2 seeds)	-2.4 to -2.1	0.049
Synapse-weighted	-1.1	0.488
Synthetic inputs (3)	-3.1 to +0.8	0.049–0.244

V. FOUR CONNECTOMES

We repeat both approaches on every hemisphere with identical, reduced settings (Table VIII, Fig. 3). Hemispheres of one animal are not independent, so we summarise by animal. The pairing is structured in 6 of 7 hemispheres (Q z from -0.0 to 17.5). Under the 2017 protocol the connectome’s difference to its null has median -1.6% (interquartile range -3.2% to -0.3%) across all 168 hemisphere–dataset–size combinations; it is negative in 134, below every null draw in 59 and above

every draw in 2; the extremes (-10.5%, +11.2%) occur at the smallest sizes, where AP is small. The mean difference is negative in 4 of 4 animals. The hemibrain is the exception to structure: by Q its pairing is indistinguishable from the null, while FlyWire’s is the most structured, so the strength of the measured structure varies between reconstructions; seven hemispheres are too few to tell whether retrieval relative to the null tracks it. On odours the primary-size difference is negative in all 7 hemispheres and all 4 animals, from -3.0% to -0.6% (3 of 7 at $p < 0.05$), and from -3.0% to -0.0% on the shared glomeruli. The even-degree control scores +1.4% to +4.2% relative to the null on average, and the fly hash leads LSH on MNIST at $k = 4$ by 2.2 – $2.9\times$ in every hemisphere. One male cannot separate sex from individual or reconstruction differences.

VI. BUDGETS

The 2017 comparison sets k active cells against k projections, which equates projection arithmetic only at $m = 20k$. The headline network has $m = 10d$ cells sampling about $d/10$ inputs each: roughly d^2 additions against $2kd$, a factor near $d/2k$ ($98\times$ on MNIST at $k = 4$; Table IX). Real-valued LSH with that arithmetic reaches 0.31–0.81 AP across the four datasets, above the fly’s best score on each, and above it at every d of the sweep (Table III). The connectome

TABLE VIII

SEVEN HEMISPHERES OF FOUR ANIMALS. CELLS, MEAN INPUTS PER CELL, FAN-OUT COEFFICIENT OF VARIATION, CO-OCCURRENCE Q AS z AGAINST $B = 100$ NULLS; ODOUR ANALYSIS AT THE PRIMARY SIZE ON EACH CONNECTOME'S DOOR GLOMERULI AND ON THE 34 ALL SHARE (RELATIVE DIFFERENCE, p , FLOOR 0.020); RANGE OVER THE 2017-PROTOCOL RUNS (4 DATASETS $\times$ 6 SIZES, $B = 50$, 5 TRIALS); EVEN-DEGREE CONTROL AND SIX-INPUT CONSTRUCTION RELATIVE TO THE NULL (MEAN OVER THOSE RUNS); FLY OVER LSH ON MNIST AT $k = 4$.

hemisphere	sex	cells	inputs	fan CV	Q z	odours $\Delta\%$ (p)	matched $\Delta\%$ (p)	protocol $\Delta\%$	even	six	fly/LSH
MaleCNS R	m	1,886	5.28	0.53	+2.8	-2.1 (0.020)	-2.6 (0.020)	-10.5 to +3.1	+3.7	+5.4	2.3
MaleCNS L	m	1,865	5.30	0.54	+5.0	-1.0 (0.257)	-0.8 (0.376)	-8.0 to +11.2	+4.2	+6.0	2.2
Hemibrain R	f	1,761	6.05	0.50	-0.0	-0.6 (0.515)	-0.0 (0.990)	-6.3 to +7.4	+2.9	+2.9	2.3
FlyWire R	f	2,406	5.00	0.47	+11.8	-2.8 (0.020)	-2.7 (0.020)	-8.9 to +4.1	+1.4	+4.2	2.4
FlyWire L	f	2,378	5.12	0.49	+17.5	-1.6 (0.099)	-1.6 (0.079)	-6.6 to +3.6	+3.0	+4.2	2.9
BANC R	f	2,046	5.03	0.57	+6.0	-3.0 (0.020)	-3.0 (0.020)	-8.9 to -0.1	+3.3	+4.7	2.3
BANC L	f	2,089	5.18	0.55	+9.1	-1.3 (0.257)	-2.2 (0.059)	-9.0 to -0.2	+1.8	+4.5	2.3

TABLE IX

PROJECTION ARITHMETIC. OPS: THE 10d FLY (10d CELLS, s INPUTS EACH) OVER $k = 4$ LSH (2kd). LSH WITH THE 10d FLY'S ARITHMETIC (REAL-VALUED PROJECTIONS) AGAINST THAT FLY'S BEST AP ($k \leq 32$).

data	d	ops	LSH proj.	LSH AP	fly 10d AP
SIFT	128	$16\times$	65	0.507	0.389
GloVe	300	$38\times$	150	0.306	0.030
MNIST	784	$98\times$	390	0.807	0.392
Odours	35	$5\times$	20	0.447	0.317

TABLE X

ODOUR ANALYSIS, MALECNS: AP OF THE FLY HASH AND OF GAUSSIAN SIGN CODES (BITS IN PARENTHESES).

k	fly	sign, k bits	storage-matched	operation-matched
8	0.197	0.019	0.264 (72)	0.305 (102)
16	0.288	0.069	0.340 (130)	0.305 (102)
32	0.353	0.151	0.394 (229)	0.305 (102)
64	0.396	0.249	0.436 (397)	0.305 (102)
92	0.406	0.296	0.454 (523)	0.305 (102)

costs $\text{nnz}(M)$ additions, the arithmetic of 98 projections; real-valued LSH with that budget beats the connectome's fly hash on 4 of 4 datasets (Fig. 1).

With one-bit codes the comparison is closer (Table X): on odours the fly hash beats sign codes of k bits ($1.37\times$ at the primary size), trails at matched storage ($0.90\times$), and at matched operations trails for small k and leads from $k = 32$ ($1.33\times$). Real-valued and one-bit baselines differ in representation and storage, so the two comparisons answer different questions. The fly hash's advantage is per active cell; per operation it holds only against one-bit codes in some settings.

VII. DISCUSSION

Across four animals the connectome behaves like a degree-preserving random matrix for similarity search: it keeps the fly hash's advantage per active cell and retrieves within a few percent of its rewirings, slightly worse where it differs, in the direction reported for odour discrimination in [5]. Its pairing is structured but does not help this task. What does matter at equal connection count is how evenly glomeruli are used: the connectome's fan-out is skewed, and evening it out improves retrieval, whereas evening out inputs per cell does not. On image and word input the random matrix of 2017 is

in that sense not only a harmless idealisation but a somewhat better hash than the circuit; on odours the difference is small. That the skew serves other ends, such as innate valence [5] or novelty detection [18], is not tested here.

Three cautions apply. The odour deficit depends on treating unmeasured DoOR entries as zero and on the glomeruli included. Image and word input is compressed to 51 dimensions and assigned to glomeruli arbitrarily. Differences between connectomes mix biology and reconstruction method.

The benchmark claim itself needs a qualifier. Its advantage is per active cell. Counted as projection arithmetic, the headline network is far more expensive than the LSH it is compared with, and real-valued projections given equal arithmetic retrieve better on every dataset and dimension we tested.

VIII. LIMITATIONS

Four animals, one male. Our reimplementations depart from the published absolute values; two protocol details come from later code, and the published score is undefined. The 2017-protocol runs across connectomes use 50 nulls and are effect estimates; inference rests on the odour tests. The odour benchmark is generated from partially observed profiles with linear mixing. The model is binary and rate-free and ignores inhibition beyond winner-take-all.

IX. CONCLUSION

The 2017 fly hashing result holds in a reimplementations and survives the substitution of four connectomes, which perform like degree-preserving random matrices, slightly worse at larger hash sizes. The circuit's uneven use of glomeruli costs retrieval; its pairing neither helps nor hurts much. The algorithm's advantage over LSH is per active cell, not per operation. Code, data checksums, results and a supplement are at <https://github.com/ssenge/FlyHash-Connectome>.

ACKNOWLEDGMENT

Claude (by Anthropic) was used as a coding and analysis assistant; the author designed the study.

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